

no idea what the weather was like over Britain. Time and again they set off in promising conditions only to encounter fog or strong head winds as they approached their targets.

Just flying the huge aircraft posed complex problems. They were more like warships than airplanes. The commander strode about his control cabin with binoculars around his neck while a coxswain steered the ship with a nautical-style wheel. Another coxswain monitored altitude and gas pressure. The engines were tended in flight by mechanics in the engine cars, and a

sailmaker checked for damage to the outer fabric. The commander and his officers were constantly engaged in complex calculations about the airship's altitude. A variety of factors made the craft rise or fall. For instance, when it rained, the water on the vast cover of the airship would increase its weight, making it lose height. Constant fine-tuning of ballast and gas pressure was needed to maintain a steady height.

But this was nothing compared to the problem of navigation. On night raids, Zeppelins often did well to find the right country, let alone a city the

size of London. The best means of navigation available was by a radio fix: the airship transmitted a signal to two ground stations, each then identified the direction of the signal, and the two direction lines allowed the airship's precise position to be calculated. But airship commanders mostly preferred to keep radio silence for fear of revealing their position to the enemy. They fell back on the age-old nautical technique of dead reckoning. If you knew how fast you had traveled, for how long, and in what direction, you could plot your position on a chart. Yet these

SHOOTING DOWN AN AIRSHIP

AT AROUND 2:30AM ON SEPTEMBER 3, 1916, a woman in north London, woken by explosions, looked out of her window and saw sailing close above "a cigar of bright silver in the full glare of about 20 magnificent searchlights." It was the Schütte-Lanz SL 11, one of 16 naval and army airships sent against London in the largest raid of the war. Also in the night sky was a B.E.2c biplane, piloted by Lieutenant William Leefe Robinson, part of the Home Defence Wing assigned to protect Britain against airship raids. Leefe Robinson and his colleagues had grown used to the unnerving experience of night flying. When an airship raid was detected, they took off



ADAPTABLE TWO-SEATER

The B.E.2c was introduced in 1914. Originally intended for reconnaissance duties, it quickly expanded to other roles, such as light bomber and Home Defence fighter.

"I saw, far behind us, a bright ball of fire... Poor fellows, they were lost the moment the ship took fire."

ZEPPELIN CAPTAIN ERNST LEHMANN
RECALLING THE LOSS OF THE SL 11

from airstrips lit by flares and, once in the air, relied on the faintly visible line of the horizon to keep their sense of balance and hold the aircraft true.

Looking for airships in total darkness was, however, a frustrating experience. Leefe Robinson had been on patrol for three hours without seeing anything when the SL 11 was suddenly lit up for all to see. Armed with explosive and incendiary ammunition, he dived toward the airship and came up under its nose, raking it with fire from bow to stern. He had shot off three drums of ammunition before he saw it begin to glow: "In a few seconds the whole rear part was blazing," he wrote. "I quickly got out of the way of the falling, blazing Zeppelin." The event was visible for miles around. Londoners cheered as the airship fell. The blaze was witnessed with very different sentiments by

Captain Ernst Lehmann,

MACABRE MEMENTOS

Pieces of the SL 11, the first airship shot down over Britain, were made into souvenirs such as these cufflinks and pin, which were sold to raise money for the Red Cross.



commander of a Zeppelin at that moment heading for home. "I saw, far behind us, a bright ball of fire," he wrote. "Poor fellows, they were lost the moment the ship took fire." The SL 11 was the first German airship to be shot down over Hertfordshire, UK. Leefe Robinson was awarded the Victoria Cross for this feat, but was shot down and taken prisoner the following year on the Western Front.

ZEPPELIN DESTROYER

The Ranken dart was designed to be dropped on an airship from above and explode after penetrating its outer cover. Explosive and incendiary ammunition eventually made airships fatally vulnerable to attack.

BURNED-OUT SHELL

The Zeppelin L 33 was one of two German airships shot down over Britain on the night of September 23-24, 1916. It was destroyed by a combination of anti-aircraft fire and air attack.

